AI and the Distribution of Prosperity:

A Unified Framework for Understanding How Technology Shapes Inequality

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August 16, 2026

Motivation

- Previous technological waves (computerization, the internet, and industrial robots) were accompanied by sharp increases in inequality.
  - Incomes at the top of the U.S. distribution grew rapidly, while incomes for the bottom half mostly stagnated or even declined (Piketty et al., 2018; Moll et al., 2022).
  - As a result, the top 1% increased its share of national income from 12% in 1978 to 20% in 2018, while the bottom 50% experienced the opposite trend (Saez and Zucman, 2019).

- How AI's gains are distributed could have profound economic, social, and political consequences for decades to come.

Questions

Through which mechanisms can AI affect income inequality?

What determines their direction and magnitude?

Setup: A Standard Decomposition of Inequality

The top group's share of total income:

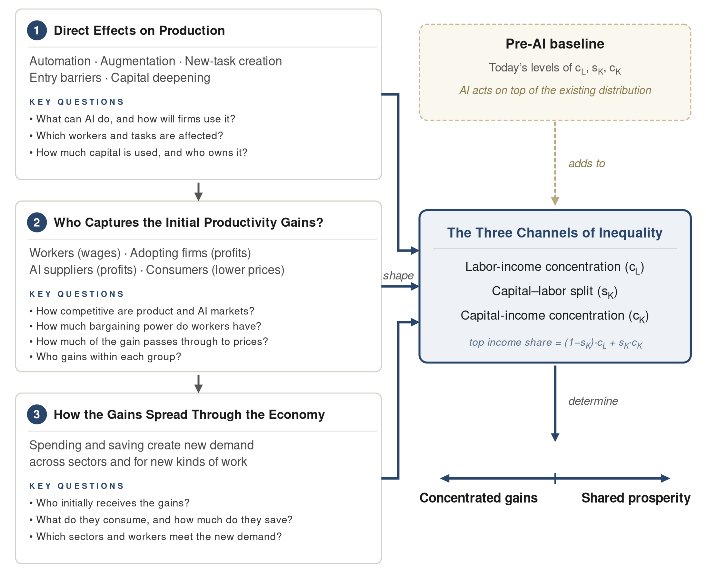
$$S_{\text{top}} \equiv (1 - s_K)\, c_L + s_K\, c_K$$

Any effect of AI on inequality must run through three channels:*

$$\Delta S_{\text{top}} = \underbrace{\Delta c_L\,(1 - s_K)}_{\text{wage concentration}} + \underbrace{\Delta c_K\, s_K}_{\text{capital concentration}} + \underbrace{\Delta s_K\,\left(c_K^{\text{postAI}} - c_L^{\text{postAI}}\right)}_{\text{shift toward capital}}$$

*Or through changes in relative prices.

This Paper: A Unified and Accessible Framework



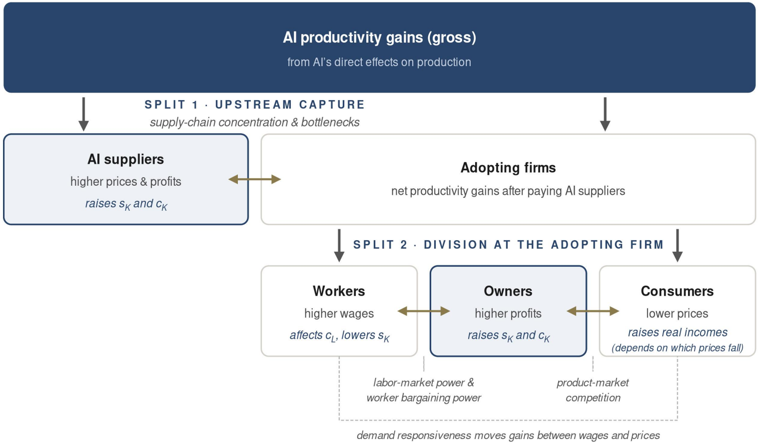
Stage 1: Direct Effects on Production

Parallels Acemoglu et al. (2026):

Mechanism	Description	The key distributional questions we discuss
Automation (§3.1.1)	AI performs existing tasks end-to-end with little human intervention, substituting capital for labor.	1. How far does automation extend? (s_K) 2. Which workers are directly affected? (c_L) 3. Where do displaced workers move? (c_L) 4. Who owns the capital that gains? (c_K) 5. Who captures the productivity gains? (c_L, s_K, c_K)
Augmentation (§3.1.2)	AI raises the productivity of workers in tasks they continue to perform.	1. Which workers experience the largest productivity gains, within and across occupations? (c_L) 2. Who captures the productivity gains? (c_L, s_K, c_K)
New-task creation (§3.1.3)	AI creates entirely new tasks and occupations.	1. How much new work is created? (s_K) 2. Who gets the new work? (c_L, s_K, c_K)
Changes in entry barriers (§3.1.4)	AI changes the expertise needed to perform tasks or enter occupations, altering occupational wage premiums.	1. Where are barriers lowered or raised? (c_L) 2. Who can access and use AI effectively? (c_L) 3. Do institutions allow boundaries to change? (c_L)
Capital deepening (§3.1.5)	AI increases the capital used per worker through AI adoption and capital-intensive AI production.	1. How much does AI increase capital intensity? (s_K) 2. Does AI substitute for or complement labor? (s_K) 3. Who owns the capital and captures its returns? (c_K)

Stage 2: Who Captures the Initial Productivity Gains?

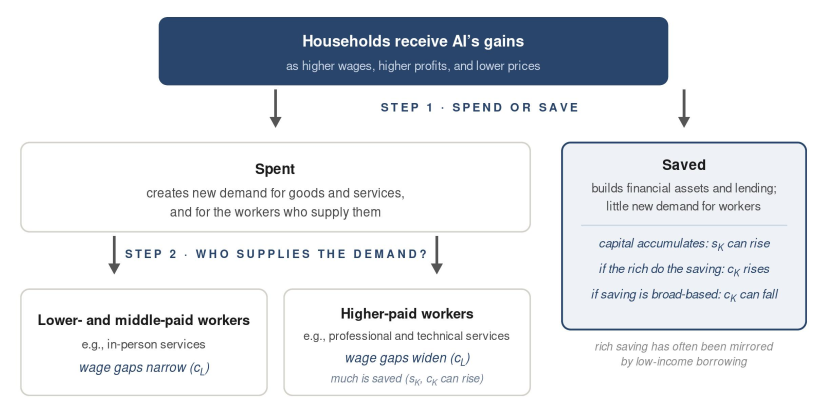
- AI suppliers, adopting firms, workers, and consumers (and within each group).
- Suppliers capture part of the gains upstream; the rest is split at adopting firms by labor-market power, product-market competition, and demand responsiveness.



Stage 3: New Demand and Where It Lands

The gains re-enter the economy:

- Spending reshapes demand for goods and services, and for the workers who provide them.
- Savings can finance investment. But when concentrated among rich households, they may instead finance borrowing by non-rich households and the government (Mian et al., 2021).



Synthesis: Two Polar Scenarios

A World of Concentrated Gains	A World of Shared Prosperity
STAGE 1 · DIRECT EFFECTS ON PRODUCTION	
• Core tasks automated in middle-paying jobs • Top workers mainly augmented • Few new tasks; mostly for specialists or AI • Higher barriers to entry-level and middle-paying work • AI capital strongly substitutes for labor	• Low- and middle-paying jobs mainly augmented • Largest gains for less-experienced and lower-skilled workers • New tasks emerge at scale and are broadly accessible • Lower barriers to high-paying professions • AI capital strongly complements labor
gains divided	*gains divided*
STAGE 2 · DISTRIBUTION OF PRODUCTIVITY GAINS	
• Supply-chain concentration and bottlenecks sustain upstream rents • Weak worker bargaining → weak wage pass-through • Weak product-market competition → weak price pass-through • Wage gains concentrated among scarce specialists • Price declines favor services for high-income households	• Competition and easing bottlenecks reduce upstream rents • Strong worker bargaining → strong wage pass-through • Strong product-market competition → strong price pass-through • Wage gains broadly shared • Cost savings reach widely consumed goods and services
gains spent or saved	*gains spent or saved*
STAGE 3 · ECONOMY-WIDE DEMAND EFFECTS	
• Gains flow mainly to owners and top earners • High saving for the rich → greater borrowing elsewhere • Spending shifts toward skill-intensive services • New demand met mostly by high-skilled workers or AI itself • Wealth accumulates among existing asset holders	• Gains broadly distributed • Stronger consumption demand • Spending supports labor-intensive, in-person services • Demand expands for middle- and lower-paid workers • Saving spread across households → broader asset ownership
$\Delta c_L > 0, \quad \Delta s_K > 0, \quad \Delta c_K > 0$ *top income share increases*	$\Delta c_L < 0, \quad \Delta s_K < 0, \quad \Delta c_K < 0$ *top income share declines*

- Not forecasts: actual outcome will combine elements of both
- Illustrate how AI can produce very different distributional outcomes
- Asymmetry: concentrated gains require continuation of recent trends; shared prosperity requires a reversal

Empirical Indicators Across the Framework

1 Direct effects on production

Automation vs. augmentation (AI usage) · breadth of adoption (surveys) · wages, employment · entry-level hiring · new tasks · skill requirements

2 Division of the gains

Supplier margins & bottlenecks · wages vs. productivity in adopters · prices & markups · demand response in exposed sectors

3 Economy-wide demand

Household spending, saving & debt · composition of spending · where employment grows

The scoreboard

The three channels (c_L, s_K, c_K) in distributional national accounts.

Existing Datasets and Measurements (+ Remaining Gaps)

What already exists:

Official statistics

- BLS: jobs, wages, prices
- CPS & expenditure surveys
- Census BTOS adoption
- Distributional accounts

New ecosystem

- Provider usage: Anthropic, OpenAI, Google
- Postings: Lightcast, Revelio
- Compute: Epoch AI
- Payroll: ADP

Early integration

- Stanford AI Indicators
- Canaries dashboard
- Linked postings & resumes

Remaining gaps:

Some Poorly Measured, Others Poorly Linked

Poorly measured: productivity gains, automation costs, new tasks, supply-chain rents, demand responsiveness, and AI-capital ownership.

Poorly linked: few datasets connect AI use and labor-market outcomes to market power, bottlenecks, ownership, and sectoral income and price elasticities.

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